

# Care Transition Efficiency and Placement Outcome Analytics: A Process-Efficiency Framework for the U.S. Unaccompanied Alien Children Program

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**Abstract**—The U.S. Unaccompanied Alien Children (UAC) Program is commonly monitored through aggregate custody counts, which describe system capacity but not the efficiency of the underlying care pipeline. This paper reframes the program’s daily public reporting data as a multi-stage flow system spanning Customs and Border Protection (CBP) custody, transfer to the Department of Health and Human Services (HHS), and discharge to a vetted sponsor. Using 720 days of daily reporting data from January 2023 to December 2025, we construct five derived process-efficiency metrics – Transfer Efficiency Ratio, Discharge Effectiveness, Pipeline Throughput, Backlog Accumulation Rate, and Outcome Stability Score – and analyze temporal patterns, bottlenecks, and a structural regime shift identified in February 2025. Results show that while total system volume fell by approximately 70% after the shift, transfer efficiency degraded by 48.9%, discharge effectiveness fell by 78.1%, and outcome stability – the consistency of placement outcomes – declined by 13.9%, indicating that reduced caseload did not translate into a more efficient or more predictable pipeline. An interactive Streamlit dashboard was developed to operationalize these metrics for ongoing monitoring. The findings suggest that process-efficiency monitoring surfaces operational risks that pure capacity counts conceal, and that efficiency degradation and volume reduction are separable phenomena requiring separate policy responses.

**Index Terms**—process analytics, care pipeline, unaccompanied minors, efficiency metrics, time series analysis, public sector data

## I. INTRODUCTION

The Unaccompanied Alien Children (UAC) Program operates as a multi-stage care and reunification pipeline rather than a static custodial system. Children apprehended at the border are first held in CBP custody, then transferred to HHS care for medical screening, sheltering, and case management, and finally discharged to a vetted sponsor, typically a family member, for reunification. From a policy and humanitarian standpoint, the speed, continuity, and reliability of movement through this pipeline are as consequential as the raw capacity of any single stage.

Public monitoring of the program has historically emphasized aggregate counts – how many children are currently in CBP custody, or currently in HHS care – because these figures are straightforward to report and interpret. However, aggregate counts answer the question of *how many*, not the question of *how well*. A custody count that remains stable over time could reflect either a well-functioning pipeline processing children at a healthy rate, or a stagnant pipeline in which intake and

discharge have both slowed in tandem. Without structured transition analytics, these two very different operational states are indistinguishable from capacity data alone, and system bottlenecks remain hidden until they manifest as visible crises.

This paper addresses that gap by treating the UAC dataset as a process to be measured rather than a population to be counted. We formalize the pipeline as three linked stages, derive five efficiency metrics from the six variables in the program’s daily public reporting, and apply time-series and structural-break analysis to characterize how the pipeline’s efficiency and stability have evolved from January 2023 through December 2025.

The remainder of this paper is organized as follows. Section II describes the dataset and preprocessing steps. Section III defines the analytical methodology, including the five key performance indicators and the bottleneck-detection procedure. Section IV presents results, including a structural regime shift identified in the data. Section V discusses the implications of these findings. Section VI describes the accompanying interactive dashboard. Section VII concludes.

## II. DATASET DESCRIPTION

The dataset consists of daily public reporting figures published for the UAC Program, covering the period from January 12, 2023 to December 21, 2025. After removing trailing empty records, 720 valid daily observations remain. Each record contains six fields, summarized in Table I.

TABLE I  
DATASET FIELDS

| Field               | Description                                 |
|---------------------|---------------------------------------------|
| Date                | Reporting date                              |
| CBP Apprehended     | Daily intake volume into CBP custody        |
| CBP Custody         | Active children in CBP custody on that date |
| CBP Transferred Out | Daily flow from CBP into HHS system         |
| HHS Care            | Active children in HHS care on that date    |
| HHS Discharged      | Daily successful sponsor placements         |

### A. Data Quality and Preprocessing

Three preprocessing steps were required before analysis. First, the HHS Care field was stored as a comma-delimited string (e.g., “2,484”) rather than a numeric type and required

parsing. Second, the Date field was stored as a full-text date string and was parsed into a standard datetime format. Third, the raw records were listed in descending chronological order and were re-sorted ascending for time-series analysis.

An inspection of the interval between consecutive reporting dates confirmed that the program reports on a business-day cadence: 558 of 719 intervals were exactly one day (consecutive weekdays), 122 intervals were three days (spanning a weekend), and 23 intervals were four days (spanning a long weekend or holiday), with a small number of irregular gaps of five or more days coinciding with holiday clusters. This cadence was accounted for explicitly in the weekday-pattern and backlog analyses described in Section III.

### III. METHODOLOGY

#### A. Pipeline Modeling

The program is modeled as a three-stage flow: CBP custody, HHS care, and sponsor placement. Daily movement between stages is captured directly by the reported transfer and discharge fields, allowing the system to be analyzed using standard flow-accounting principles: a stage's population on a given day is a function of its population on the prior day plus inflow minus outflow.

#### B. Key Performance Indicators

Three KPIs were specified directly by the program's process-efficiency framework, and are computed as ratios of the raw reported fields:

$$\text{Transfer Efficiency Ratio} = \frac{\text{CBP Transferred Out}}{\text{CBP Custody}} \quad (1)$$

$$\text{Discharge Effectiveness} = \frac{\text{HHS Discharged}}{\text{HHS Care}} \quad (2)$$

$$\text{Pipeline Throughput} = \frac{\text{Total Exits}}{\text{Total Entries}} \quad (3)$$

where Total Exits is the sum of CBP Transferred Out and HHS Discharged, and Total Entries is the daily CBP Apprehended count. A throughput value above 1.0 indicates the pipeline is clearing more cases than it is admitting on a given day.

Two additional KPIs were not explicitly formulaic in the source specification and were operationalized as follows. The **Backlog Accumulation Rate** is defined as the seven-day rolling average of daily net flow, where net flow is CBP Apprehended minus HHS Discharged:

$$\text{Net Flow}_t = \text{Apprehended}_t - \text{Discharged}_t \quad (4)$$

$$\text{Backlog Rate}_t = \frac{1}{7} \sum_{i=t-6}^t \text{Net Flow}_i \quad (5)$$

A sustained positive value indicates the pipeline is accumulating cases faster than it discharges them; a sustained negative value indicates the backlog is being cleared.

The **Outcome Stability Score** quantifies the day-to-day consistency of discharge performance using a 14-day rolling coefficient of variation of Discharge Effectiveness, inverted so that higher values indicate greater stability:

$$\text{CV}_t = \frac{\sigma_{14}(\text{Discharge Effectiveness})_t}{\mu_{14}(\text{Discharge Effectiveness})_t} \quad (6)$$

$$\text{Stability Score}_t = \frac{1}{1 + \text{CV}_t} \quad (7)$$

#### C. Bottleneck Detection

A bottleneck period is defined as a run of five or more consecutive days in which net flow is strictly positive, i.e., intake sustained exceeds discharge for a period long enough to indicate accumulating backlog rather than single-day noise. Consecutive-day runs were identified using a streak-labeling procedure applied to the sign of daily net flow.

#### D. Regime Segmentation

Exploratory analysis of monthly HHS Care averages revealed a sharp discontinuity beginning in January-February 2025, with the average HHS care load falling from 6,419 in December 2024 to 2,782 in February 2025 and stabilizing near 2,000–2,400 for the remainder of the observed period. The dataset was accordingly split into two regimes – *pre-shift* (before February 1, 2025) and *post-shift* (on or after February 1, 2025) – so that subsequent stability and backlog metrics could be interpreted in the correct operational context rather than conflated across two structurally different periods.

### IV. RESULTS

#### A. Pipeline Flow and the February 2025 Regime Shift

Fig. 1 shows HHS care load and CBP custody over the full observation window. All five raw volume metrics – apprehensions, CBP custody, transfers, HHS care load, and discharges – exhibit a common, near-simultaneous decline beginning in January 2025. Table II summarizes the magnitude of this shift across the core metrics.

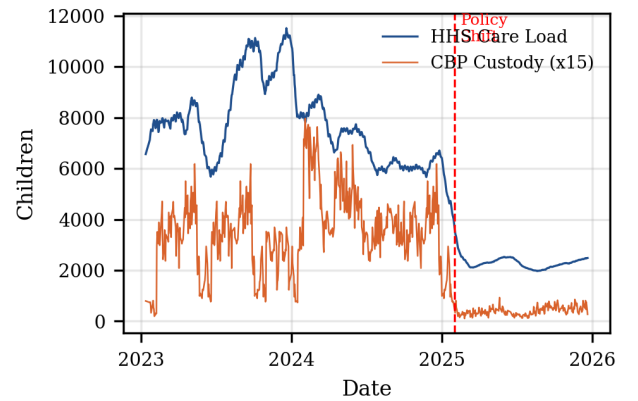


Fig. 1. HHS care load and CBP custody (scaled) over the observation period. The dashed line marks the identified February 2025 regime shift.

TABLE II  
PRE- VS. POST-SHIFT METRIC COMPARISON

| Metric                  | Pre-2025 | Post-2025 | $\Delta$ |
|-------------------------|----------|-----------|----------|
| HHS Care Load (avg.)    | 7,686    | 2,295     | −70.1%   |
| CBP Custody (avg.)      | 234      | 27        | −88.4%   |
| Transfer Efficiency     | 0.811    | 0.414     | −48.9%   |
| Discharge Effectiveness | 0.031    | 0.007     | −78.1%   |
| Pipeline Throughput     | 3.95x    | 4.09x     | +3.6%    |
| Outcome Stability       | 0.817    | 0.703     | −13.9%   |

Notably, while overall system volume contracted sharply, Pipeline Throughput remained essentially unchanged, since both entries and exits fell together. Transfer Efficiency and Discharge Effectiveness, by contrast, degraded substantially – indicating that the *rate* at which children move through each stage slowed even as the *absolute number* of children in the system shrank.

### B. Monthly KPI Trends

Fig. 2 presents monthly averages for four KPIs across the full observation window. Transfer Efficiency and Discharge Effectiveness both show a stable, gently fluctuating pattern through 2023 and 2024, followed by an abrupt step-decline at the regime shift with no recovery through the remainder of 2025. The Backlog Accumulation Rate, conversely, moves from a persistently negative regime pre-shift, indicating the pipeline was net-clearing cases throughout 2023–2024, to a near-zero flat regime post-shift, consistent with a smaller system in rough equilibrium rather than active backlog clearance.

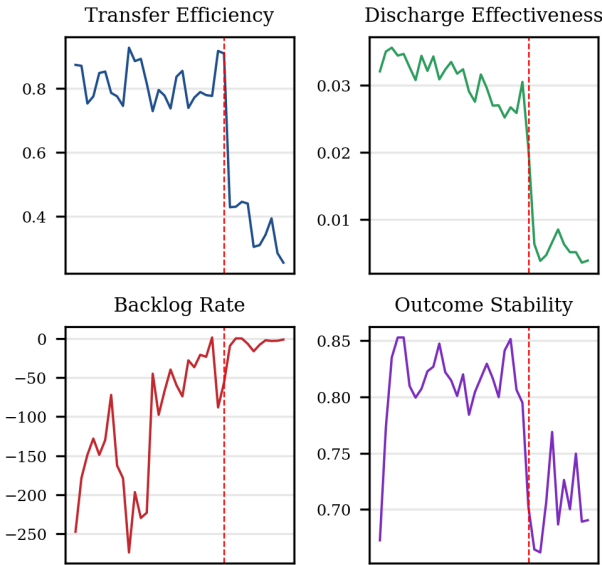


Fig. 2. Monthly trends for four core KPIs. Dashed line marks the February 2025 regime shift.

### C. Bottleneck Analysis

Applying the five-day sustained-imbalance criterion described in Section III-C, exactly one bottleneck period was

detected across the full 720-day window: March 30 to April 3, 2025, five consecutive days with a cumulative net excess of 22 children. This period falls within the post-shift regime, meaning that despite substantially lower system volume, the only detected sustained backlog occurred after the shift – suggesting that the reduced-capacity operating environment introduced by the shift is more fragile to short-term imbalances than the higher-volume pre-shift environment, which was more consistently net-clearing.

### D. Temporal Patterns

Discharge Effectiveness by day of week (Fig. 3) shows zero recorded activity on Saturdays, confirming the program’s reporting and operational cadence excludes Saturday processing. Sunday shows the highest average Discharge Effectiveness (0.0284), followed by Thursday (0.0279), while Tuesday shows the lowest (0.0187). The elevated Sunday figure is consistent with a catch-up effect, in which cases held over the weekend are processed as the work-week resumes.

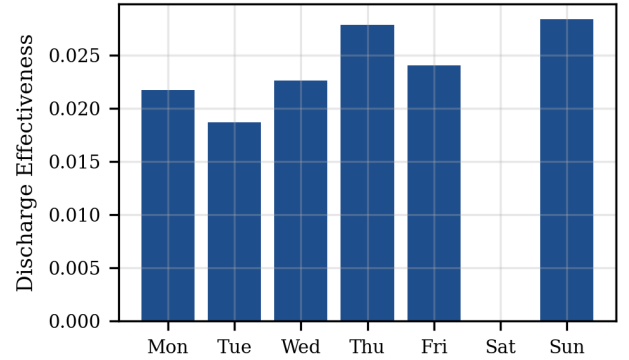


Fig. 3. Average Discharge Effectiveness by day of week. Saturday shows no recorded activity.

### E. Outcome Stability

Fig. 4 shows the Outcome Stability Score across the full period. Pre-shift, the score remains within a comparatively narrow band of approximately 0.80–0.85. Post-shift, the score becomes markedly more volatile, ranging from approximately 0.55 to 0.85, with several sharp dips. This is a counter-intuitive result: a smaller system, with less caseload pressure, exhibits *less* predictable discharge performance, not more. This pattern points toward an operational or process-level source of variability rather than a purely volume-driven one.

## V. DISCUSSION

The central finding of this analysis is that volume and efficiency are separable dimensions of pipeline performance, and that the February 2025 regime shift affected them differently. A capacity-only monitoring approach, tracking only the number of children currently in custody or care, would report the post-shift period as an unambiguous improvement, since every raw count declined substantially. The efficiency-oriented view developed in this paper complicates that reading: transfer

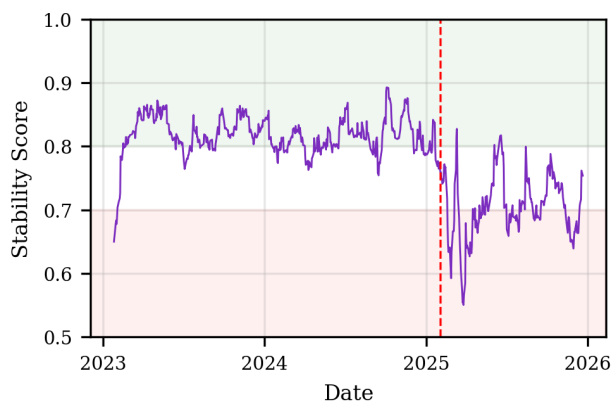


Fig. 4. Outcome Stability Score over time, with stable ( $\geq 0.8$ ) and volatile ( $< 0.7$ ) reference bands. Post-shift variability is visibly higher.

speed, discharge speed, and outcome consistency all declined even as the system shrank, and the only sustained bottleneck detected in the entire 720-day window occurred in the smaller, post-shift system.

Two explanations are consistent with this pattern and merit further investigation with data beyond the scope of this dataset. First, a smaller intake volume may not have been matched by a proportional reduction in staffing or fixed processing steps, so that per-case friction increased even as aggregate load fell. Second, changes to intake composition, screening requirements, or sponsor-vetting procedures coincident with the regime shift could independently slow each case’s transition without any change in the number of cases being processed. Distinguishing between these (and other) explanations would require case-level or policy-level data not present in the aggregate daily reporting analyzed here.

From a monitoring standpoint, this analysis illustrates the value of deriving process metrics from stock-and-flow public data even when case-level detail is unavailable. The five KPIs used here required no additional data collection, only a reframing of six already-published daily figures as a pipeline rather than a set of independent counts.

## VI. DASHBOARD IMPLEMENTATION

To operationalize the metrics described above for ongoing monitoring, an interactive dashboard was built using Streamlit and Plotly. The dashboard provides: (1) a live pipeline flow visualization spanning all three stages; (2) toggleable panels for each of the five KPIs with monthly trend views; (3) an automated bottleneck-detection panel that surfaces any currently active or historical five-plus-day imbalance periods; (4) an outcome-stability view with visual stable/volatile reference bands; and (5) date-range and policy-regime filters allowing a user to isolate any sub-period of interest. The dashboard is designed to allow a non-technical stakeholder to answer the four questions posed in the original problem statement – transfer speed, discharge pace, bottleneck location,

and outcome trend – without needing to interact with the underlying data directly.

## VII. CONCLUSION

This paper reframed nearly three years of UAC Program daily reporting data from a capacity-monitoring lens to a process-efficiency lens, deriving five KPIs – Transfer Efficiency Ratio, Discharge Effectiveness, Pipeline Throughput, Backlog Accumulation Rate, and Outcome Stability Score – from the program’s six published fields. The analysis identified a structural regime shift beginning in February 2025 in which total system volume fell by roughly 70%, but transfer and discharge efficiency degraded by 48.9% and 78.1% respectively, and outcome stability declined by 13.9%. These findings demonstrate that reduced caseload and improved pipeline efficiency are not the same phenomenon, and that process-efficiency monitoring of the type developed here can surface operational risks – such as slowing transitions and increasing outcome volatility – that remain invisible to aggregate capacity counts alone. An accompanying interactive dashboard operationalizes these metrics for ongoing use by program stakeholders.

## ACKNOWLEDGMENT

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